

EDA using Univariate Analysis

What is Univariate Analysis?

Univariate analysis involves isolating a single attribute (column) from your dataset and examining its structural properties independently. The primary objective is to map out frequencies, ranges, and statistical shapes without looking at relationships between variables.

1. Analyzing Categorical Features

Categorical attributes capture labels, groups, or discrete rankings. Univariate analysis for these columns focuses on calculating frequencies and relative percentages using tools like frequency counts, count plots, and pie charts.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('titanic.csv')

# Frequency count of distinct category groups
print(df['Survived'].value_counts())

# Generating a visual Frequency Distribution using Seaborn
sns.countplot(data=df, x='Survived')
plt.show()

# Relative Percentage Breakdown using Matplotlib Pie Chart
df['Pclass'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.show()
```

2. Analyzing Numerical Features: Continuous Distributions

Numerical data features are composed of continuous ranges. To understand how values span across your dataset, you can group them into numeric buckets using Histograms or trace continuous distribution trends using Probability Density Functions (PDF) or Distribution Plots.

```
# Visualizing frequencies across grouped numeric buckets
plt.hist(df['Age'], bins=20)
plt.show()

# Tracing data distribution via Kernel Density Estimation (KDE)
sns.distplot(df['Age'], hist=True)
```

3. Mathematical Outlier Auditing: Box Plots

Outliers can severely distort predictive machine learning algorithms. Box plots provide a structural five-number summary (minimum, 25th percentile, median, 75th percentile, and maximum) to help visually isolate and manage value anomalies.

```
# Visualizing Interquartile Ranges (IQR) and isolating mathematical anomalies
sns.boxplot(x=df['Fare'])
plt.show()
```

4. Evaluating Asymmetry: Skewness Measurements

Data balance heavily impacts algorithm stability. Calculating the third statistical moment (skewness coefficient) helps you mathematically verify if your features form a symmetrical distribution or if they tilt unevenly toward a specific side.

```
# Calculating distribution tilt: 0 = symmetrical, >0 = right-skewed, <0 = left-
print("Feature Skewness Metric:", df['Age'].skew())
```

Key Developer Rule: Always identify the data type of a column before running univariate analysis. Categorical columns are best examined using counts and percentages, while numerical features should be checked for distribution shape and outliers before training models.